9.49. Isolation by Disruptive Selection. J. B. GIBSON and J. M. THODAY (Cambridge, Great Britain).

Thoday and Gibson[1] exposed a population "Southacre" of *Drosophila melanogaster* to disruptive selection for sternoplural chaeta number using a breeding system that ensured that all selected flies were together in a single 3 × 1 in. vessel for the limited period during which all mating had to occur. Their population rapidly split into two components between which little if any effective mating occurred.

Selection was discarded in this line when extreme fertility trouble occurred at generation 28. However, isolated high and low chaeta number lines were maintained. A new line was started from the "Southacre" stock under the same selection régime. Divergence was more rapid in the second experiment than in the first and the two halves of the population became completely distinct by the seventh generation.

Mating preference tests in this new line using chaeta number as a marker have indicated an isolation index of +0.54 (where 0 would represent random mating and +1 complete positive assortative mating). There is therefore evidence of strong mating preferences in the line. It does not however seem sufficiently strong to provide a complete explanation of the lack of hybrids in the line itself. Furthei evidence suggests that the hybrids that are formed may compete poorly with the non-hybrids.

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1. *Nature* **193,** 1962.